

PROBLEM 15.18 Working drawing (in.)

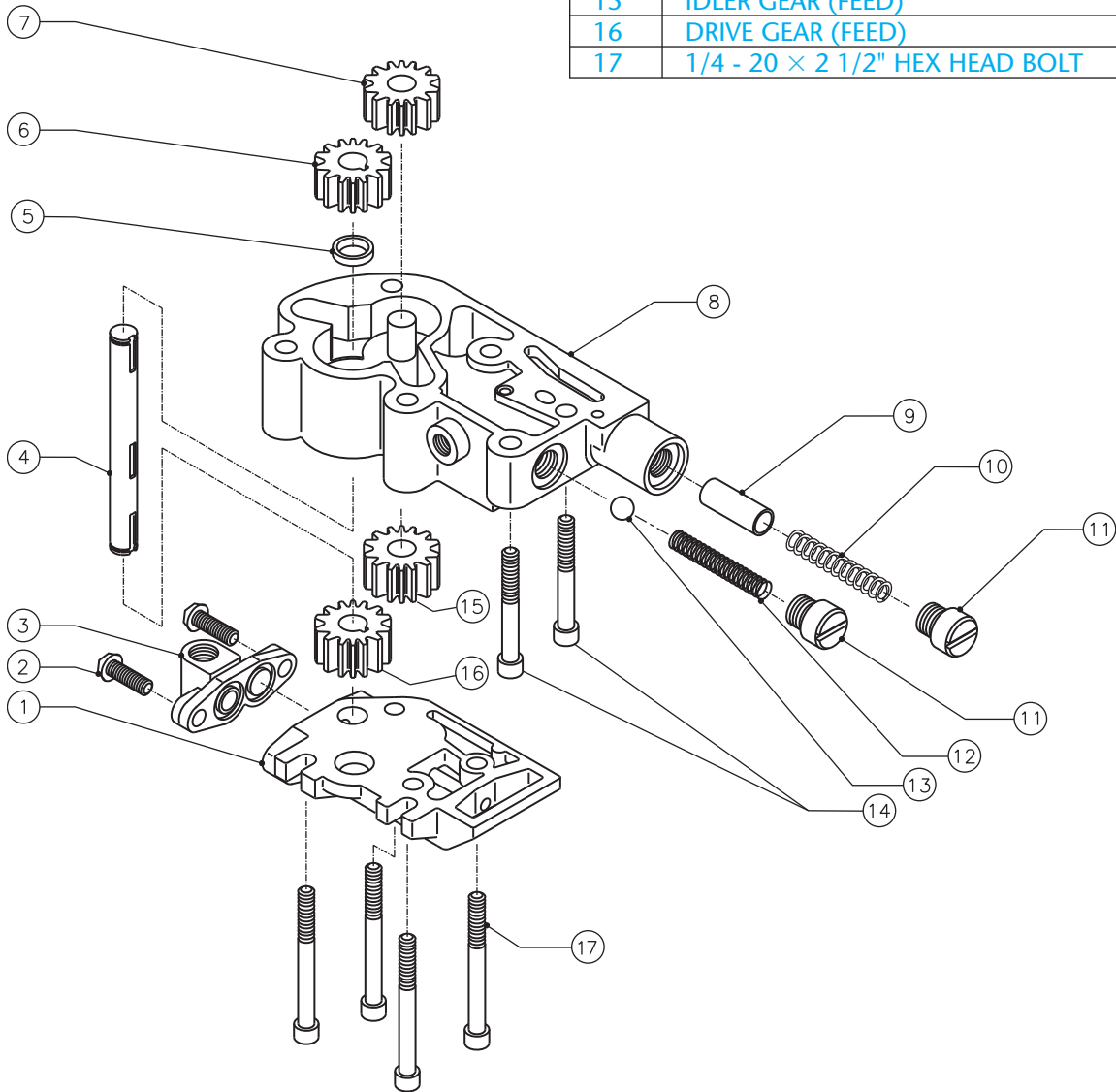
Assembly Name: Oil Pump

SPECIFIC INSTRUCTIONS: This problem is challenging, and contains applications that include gears and seals. You should study gears and related drafting practices covered in Chapter 16 before completing this problem.

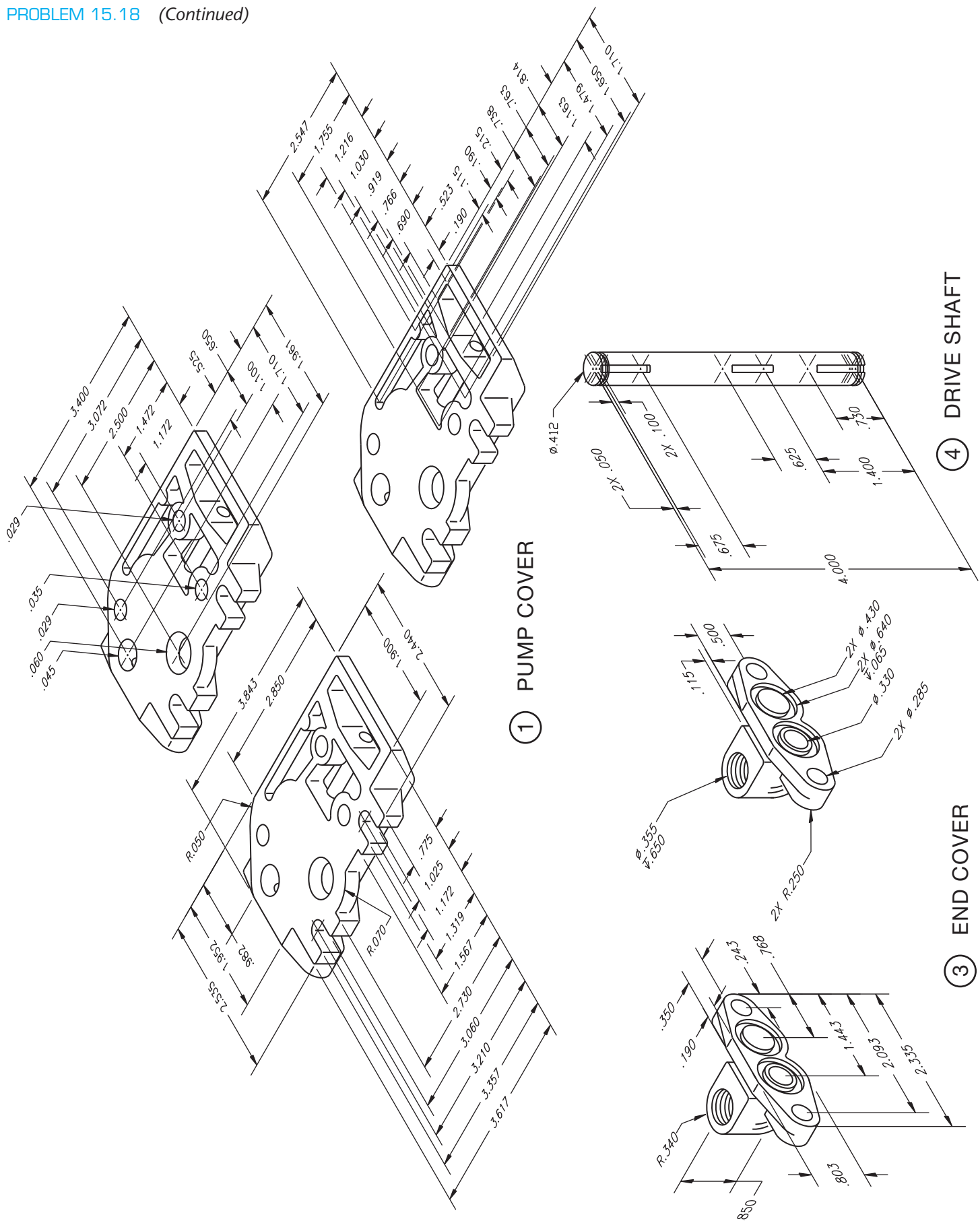
Prepare a complete set of working drawings with one detail drawing per sheet and the assembly and parts list on another sheet. When preparing the assembly drawing, use separate balloons for each part in each view or use only one balloon in the view that most clearly identifies the part. This is an engineering prototype project. Errors are likely in this problem. Verify dimensions during assembly. Establish tolerances between mating parts. There

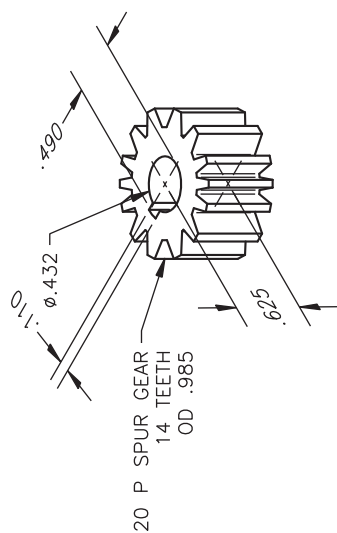
are several purchase parts in this assembly. Manufacturer research is needed to locate available products.

PARTS LIST		
ITEM	DESCRIPTION	QTY
1	PUMP COVER	1
2	1/4 - 20 × 3/4" BUTTON HEAD BOLT	2
3	END COVER	1
4	DRIVE SHAFT	1
5	OIL SEAL	1
6	DRIVE GEAR (RETURN)	1
7	IDLER GEAR (RETURN)	1
8	OIL PUMP BODY	1
9	PLUNGER VALVE	1
10	PLUNGER SPRING	1
11	END CAP	2
12	PSI CONTROL SPRING	1
13	CHECK BALL	1
14	1/4 - 20 × 2" HEX HEAD BOLT	2
15	IDLER GEAR (FEED)	1
16	DRIVE GEAR (FEED)	1
17	1/4 - 20 × 2 1/2" HEX HEAD BOLT	4

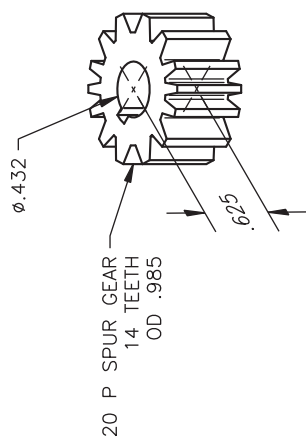


PROBLEM 15.18 (Continued)

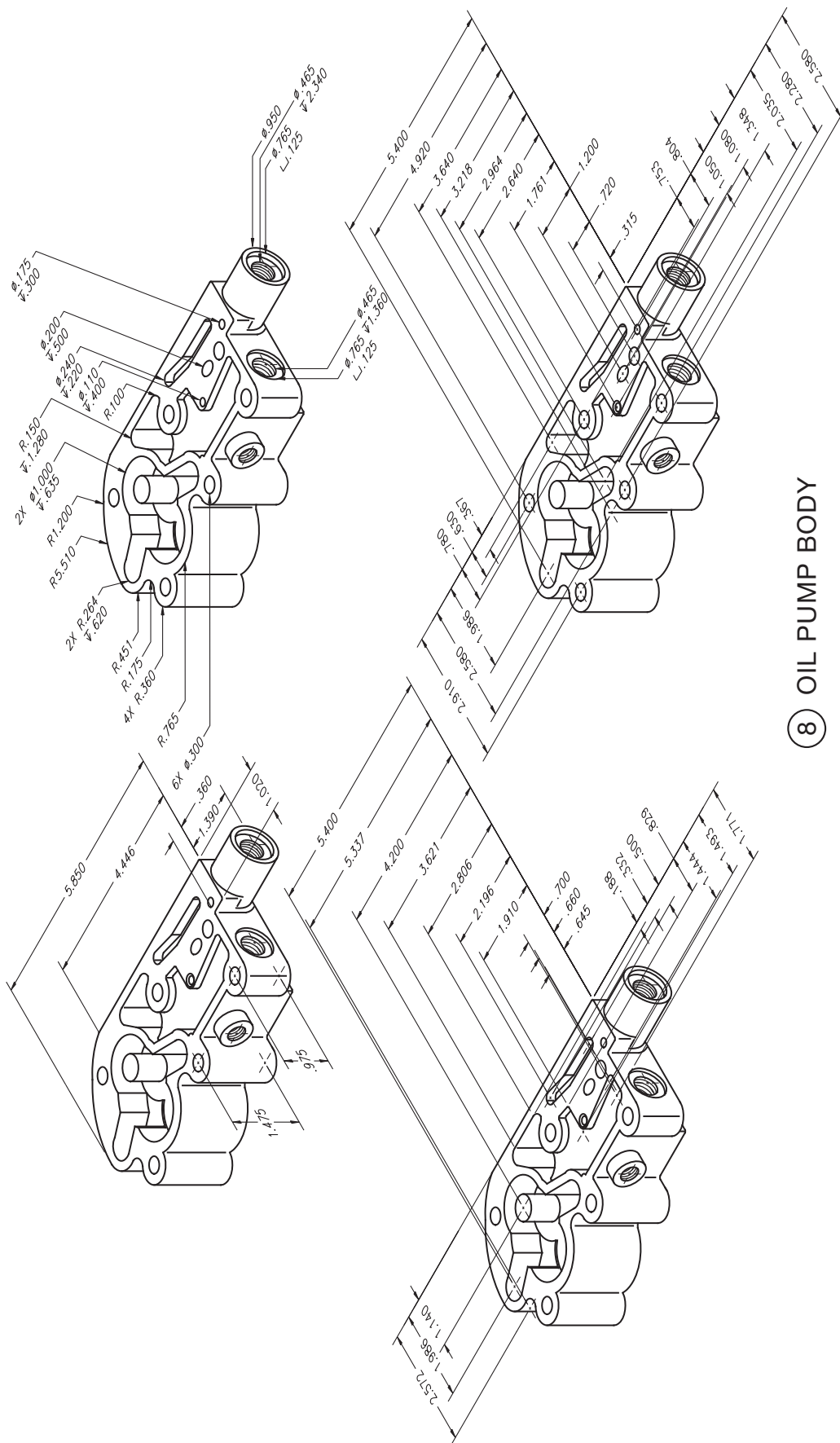




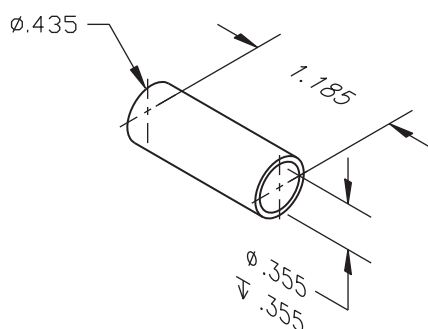
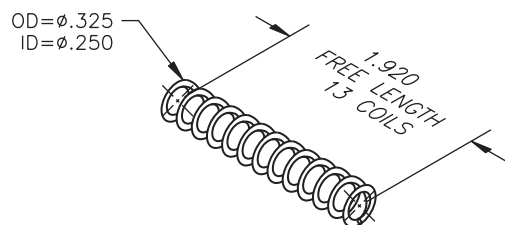
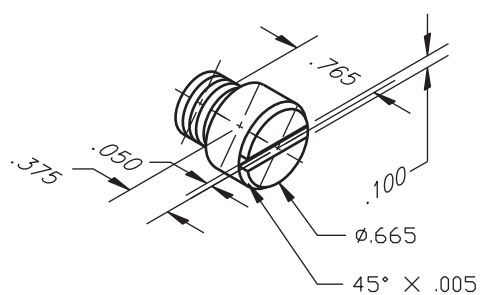
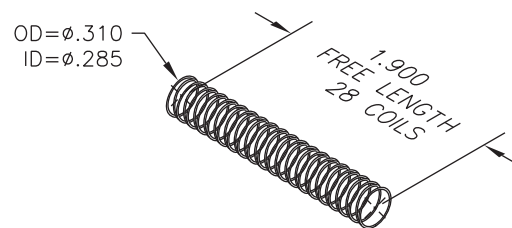
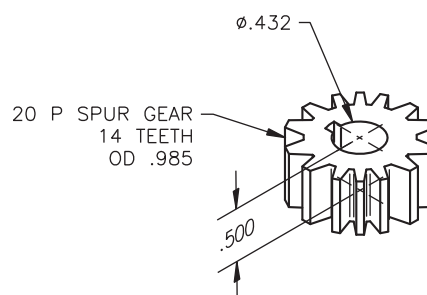
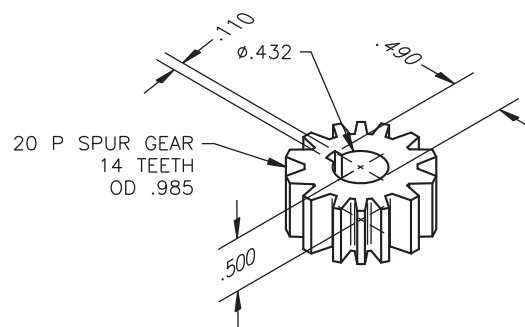
⑥ DRIVE GEAR (RETURN)



⑦ IDLER GEAR (RETURN)



⑧ OIL PUMP BODY

PROBLEM 15.18 (Continued)

9 PLUNGER VALVE

10 PLUNGER SPRING

11 END CAP

12 PSI CONTROL SPRING

15 IDLER GEAR (FEED)

16 DRIVE GEAR (FEED)